

Modern Hopfield Neural Network

Ref: [arxiv:2008.02217](https://arxiv.org/abs/2008.02217)

Consider K patterns $\vec{\xi}_1, \vec{\xi}_2, \dots, \vec{\xi}_K$ stored in the modern Hopfield neural network of N neurons.

$$\vec{\xi}_i \in \mathbb{R}^N \quad \text{with } i=1, 2, \dots, K$$

These patterns form a matrix

$$\Xi = (\vec{\xi}_1, \vec{\xi}_2, \dots, \vec{\xi}_K) \in \mathbb{R}^{N \times K}$$

The largest pattern index $M = \max_i \|\vec{\xi}_i\|$

where the Euclidean norm (L^2 norm) of a vector

$$\vec{v} = (v_1, v_2, \dots, v_N)$$

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_N^2}$$

The neuron values (state vector) are represented with

$$\vec{x} \in \mathbb{R}^N.$$

The energy of the modern Hopfield neural network is

$$E = -\text{lse}(\beta, \Xi^T \vec{x}) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \log N + \frac{1}{2} M^2$$

$$= -\beta^{-1} \ln \left(\sum_{i=1}^K \exp(\beta (\Xi_i^T \vec{x})) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \log N + \frac{1}{2} M^2$$

where $\uparrow$

- $\beta > 0$: effective temperature

$$\bullet \Xi^T \vec{x} = \begin{pmatrix} \vec{z}_1^T \\ \vec{z}_2^T \\ \vdots \\ \vec{z}_K^T \end{pmatrix} \vec{x} = \begin{pmatrix} \vec{z}_1^T \cdot \vec{x} \\ \vec{z}_2^T \cdot \vec{x} \\ \vdots \\ \vec{z}_K^T \cdot \vec{x} \end{pmatrix}$$

- each element is a dot product. It represents the similarity between the i -th pattern ($\vec{z}_i$) and the current state vector $\vec{x}$ of the network
- If the current state is very similar to pattern $\vec{z}_i$, the dot product will be a very large positive value $\rightarrow$ high alignment
- If they are unrelated, the dot product will be nearly zero.
- If the state vector $\vec{x}$ is exactly opposite of a pattern, the dot product will be a large negative value.
- In the original Hopfield neural network

$$E = -\frac{1}{2} \sum_{i=1}^K (\vec{z}_i^T \vec{x})^2$$

The energy is minimized whenever the current state vector $\vec{x}$ aligns closely with any of the stored patterns $\vec{z}_i^T$. This

creates "valleys" in the energy landscape, effectively turning each stored pattern into a local minimum (an attractor) that the neurons will naturally settle into.

- The modern Hopfield neuron network generalized the energy expression so that it can store more patterns and the pattern vectors and state vector can be continuous valued.
- The state vector is updated following an update rule to search for a pre-stored pattern and hence minimizes the energy (though it may only find the local not global minimum).
- The update rule for the modern Hopfield neural network is

$$\vec{x}^{\text{new}} = \underset{\substack{\downarrow \\ N \times K}}{\Xi} \text{softmax} \left(\beta \underset{\substack{\downarrow \\ K \times N}}{\Xi}^T \underset{N \times 1}{\vec{x}} \right)$$

- The softmax function of vector $\vec{v} = (v_1, v_2, \dots, v_k)$

$$\text{softmax}(\vec{v})_i = \frac{e^{v_i}}{\sum_{j=1}^k e^{v_j}}$$

- This update rule acts like an "attention mechanism"

$\Xi^T \vec{x}$: the dot product between every stored pattern in Ξ and the current state vector $\vec{x}^{\text{old}}$. It gives a vector of "scores" representing how much the current state "recognizes" each pattern.

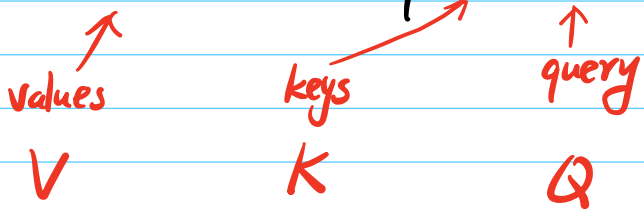
$\text{softmax}(\beta \Xi^T \vec{x})$: It turns these dot products into a probability distribution (weights that sum to 1).
If one pattern has a much higher dot product than the others, its weight will be close to 1.

β : the inverse temperature ($\beta \equiv \frac{1}{k_B T}$) controls how sharp it is. High β makes it pick only the best match; low β makes it a blurry average of several matches.

$\Xi \cdot \text{weights}$: weighted average of all stored patterns.
(or retrieval)
The network is literally pulling the state $\vec{x}$ toward the most relevant patterns in the memory matrix.

This specific update rule is mathematically identical to the **Large Language Model (Transformer)** Attention mechanism.

$$\vec{x}^{\text{new}} = \Xi \operatorname{softmax}(\beta \Xi^T \vec{x})$$



values keys query
V K Q

$$\text{Attention}(Q, K, V) = \operatorname{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Imagine you are in a massive digital library (like Chat GPT's brain)

- Query (Q) : This is the search term. It's the words the AI is currently processing (e.g. The cat sat on the ...)
It is looking for what comes next
- Keys (K) : These are the book labels. Every stored pattern or word has a "label" that describes what it's about.
- Value (V) : This is the actual content inside the books.

Note: In modern Hopfield neural network, the label and the content are the same thing.

- The "Aha" moment

ChatGPT doesn't just remember one thing at a time. Because of this equation, it can superpose (blend) memories. If the Query matches two different patterns slightly, the softmax allows it to take a little bit of information from both to form a new, creative thought.

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a+x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a+x}} = x$, let's start by squaring both ...

$$(\sqrt{a - \sqrt{a+x}})^2 = x^2 \implies a - \sqrt{a+x} = x^2.$$

Rearrange to isolate the inner square root term:

$$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$$

...

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be ...

We started with the equation:

$$\sqrt{a - \sqrt{a+x}} = x$$

First, let's square both sides:

$$a - \sqrt{a+x} = x^2 \implies \sqrt{a+x} = a - x^2$$

Next, I could square both sides again, treating the equation: ...

...

Table 3 | An interesting "aha moment" of an intermediate version of DeepSeek-R1-Zero. The model learns to rethink using an anthropomorphic tone. This is also an aha moment for us, allowing us to witness the power and beauty of reinforcement learning.

A few technical aspects of modern Hopfield neural networks

§1. $Z_q(15)$, an alternative form of energy

$$E = -\ln(\beta, \Xi^T \vec{x}) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln K + \frac{1}{2} M^2$$

$$= -\beta^{-1} \ln \left(\sum_{i=1}^K \exp(\beta \Xi_i^T \vec{x}) \right) + \beta^{-1} \ln K + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2$$

$$\stackrel{!}{=} -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp \left(-\frac{1}{2} \beta (M^2 - \|\vec{\xi}_i\|^2) \right) \exp \left(-\frac{1}{2} \beta \|\vec{\xi}_i - \vec{x}\|^2 \right) \right)$$

This is $\vec{\xi}_i$

Because

$$a = -\ln(e^{-a}) = -\beta^{-1} \ln(e^{-\beta a})$$

$$a - \beta^{-1} \ln S = -\beta^{-1} \ln(e^{-\beta a}) - \beta^{-1} \ln S = -\beta^{-1} \ln(S e^{-\beta a})$$

$$\text{let } a = \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2 + \beta^{-1} \ln K$$

$$S = \sum_{i=1}^K \exp(\beta \Xi_i^T \vec{x})$$

we find

$$E = -\beta^{-1} \ln \left(\sum_{i=1}^K \exp(\beta \Xi_i^T \vec{x}) \exp \left(-\frac{\beta}{2} \vec{x}^T \vec{x} - \frac{\beta}{2} M^2 - \ln K \right) \right)$$

$$\text{since } \exp(-\ln K) = \frac{1}{e^{\ln K}} = \frac{1}{K}$$

Hence

$$E = -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp(\beta \Xi_i^T \vec{x}) \exp \left(-\frac{\beta}{2} \vec{x}^T \vec{x} - \frac{\beta}{2} M^2 \right) \right)$$

$$= -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp \left(\underbrace{\beta \vec{\xi}_i^T \vec{x} - \frac{\beta}{2} \|\vec{x}\|^2 - \frac{\beta}{2} M^2}_{\text{highlighted in blue}} \right) \right)$$

The term highlighted in blue can be rearranged using identity

$$\|\vec{\xi}_i - \vec{x}\|^2 = \|\vec{\xi}_i\|^2 + \|\vec{x}\|^2 - 2\vec{\xi}_i^T \vec{x}$$

$$\Rightarrow \underbrace{\beta \vec{\xi}_i^T \vec{x} - \frac{\beta}{2} \|\vec{x}\|^2}_{\text{highlighted in blue}} = \frac{\beta}{2} \|\vec{\xi}_i\|^2 - \frac{\beta}{2} \|\vec{\xi}_i - \vec{x}\|^2$$

$$\Rightarrow \beta \vec{\xi}_i^T \vec{x} - \frac{\beta}{2} \|\vec{x}\|^2 - \frac{\beta}{2} M^2 = -\frac{\beta}{2} (M^2 - \|\vec{\xi}_i\|^2) - \frac{\beta}{2} \|\vec{\xi}_i - \vec{x}\|^2$$

So

$$\exp \left(\beta \vec{\xi}_i^T \vec{x} - \frac{\beta}{2} \|\vec{x}\|^2 - \frac{\beta}{2} M^2 \right) = \exp \left[-\frac{\beta}{2} (M^2 - \|\vec{\xi}_i\|^2) \right] \exp \left[-\frac{\beta}{2} \|\vec{\xi}_i - \vec{x}\|^2 \right]$$

Therefore,

$$E = -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp \left[-\frac{\beta}{2} (M^2 - \|\vec{\xi}_i\|^2) \right] \exp \left[-\frac{\beta}{2} \|\vec{\xi}_i - \vec{x}\|^2 \right] \right)$$

Note: this expression is useful to derive some properties of the energy. (Lemma A1)

- The lower bound of E is zero

Assume the pattern most similar to the state vector $\vec{x}$ is $\vec{z}_m$, then

$$m = \arg \max_i \vec{z}_i^T \vec{x}$$

$$E = -\beta^{-1} \ln \left(\sum_{i=1}^K \exp(\beta \vec{z}_i^T \vec{x}) \right) + \beta^{-1} \ln K + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2$$

$$= -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp(\beta \vec{z}_i^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2$$

M is the largest length of the patterns

$$\geq -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp(\beta \vec{z}_i^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} \vec{z}_m^T \vec{z}_m$$

↓ the largest overlap

$$\geq -\beta^{-1} \ln \left(\frac{1}{K} \sum_{i=1}^K \exp(\beta \vec{z}_m^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} \vec{z}_m^T \vec{z}_m$$

$$= -\beta^{-1} \ln \left(\exp(\beta \vec{z}_m^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} \vec{z}_m^T \vec{z}_m$$

$$= -\vec{z}_m^T \vec{x} + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} \vec{z}_m^T \vec{z}_m$$

$$= \frac{1}{2} (\vec{z}_m - \vec{x})^T (\vec{z}_m - \vec{x})$$

$$= \frac{1}{2} \|\vec{z}_m - \vec{x}\|^2$$

$$\geq 0$$

The lower bound is attained if all patterns $\vec{z}_i$ are equal and equal to state vector $\vec{x}$

- The upper bound of E : $E \leq \beta^{-1} \ln K + \frac{1}{2} M^2$
 $E \leq 2M^2$

consider the state vector is a superposition of stored patterns

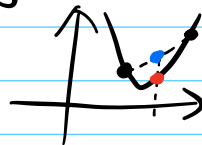
$$\vec{x} = \sum_{i=1}^k p_i \vec{z}_i \quad \sum_{i=1}^k p_i = 1 \quad \forall_i p_i \geq 0$$

(This is the so-called simplex) $\Rightarrow \vec{x}^T \vec{x} = \sum_{i=1}^k p_i \vec{z}_i^T \vec{x}$

$$E = -\beta^{-1} \ln \left(\sum_{i=1}^k \exp(\beta \vec{z}_i^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln K + \frac{1}{2} M^2$$

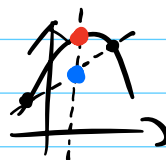
We need to apply Jensen's inequality

For a convex function f



$$f\left(\sum_i p_i x_i\right) \leq \sum_i p_i f(x_i) \quad (\text{where } p_i \geq 0, \sum_i p_i = 1)$$

For a concave function f



$$f\left(\sum_i p_i x_i\right) \geq \sum_i p_i f(x_i)$$

exp convex

sum preserves convexity

ln concave

Thus, using Jensen's inequality

$$\ln \left(\sum_{i=1}^k \exp(\beta \vec{z}_i^T \vec{x}) \right) = \ln \left(\sum_i p_i \frac{e^{\beta \vec{z}_i^T \vec{x}}}{p_i} \right)$$

$$\geq \sum_i p_i \ln \left(\frac{e^{\beta \vec{z}_i^T \vec{x}}}{p_i} \right)$$

$$= \sum_i p_i (\beta \vec{z}_i^T \vec{x} - \ln p_i)$$

$$= \beta \sum_i p_i (\vec{z}_i^T \vec{x}) - \sum_i p_i \ln p_i$$

Thus,

$$E = -\beta^{-1} \ln \left(\sum_{i=1}^k \exp(\beta \vec{z}_i^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

$$\leq - \sum_{i=1}^k p_i (\vec{z}_i^T \vec{x}) + \underbrace{\sum_i p_i \ln p_i}_{\text{dropped}} + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

$$\leq - \vec{x}^T \vec{x} + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

$$= - \underbrace{\frac{1}{2} \vec{x}^T \vec{x}}_{\text{positive}} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

$$\leq \beta^{-1} \ln k + \frac{1}{2} M^2$$

The second upper bound

$$E = -\beta^{-1}/n \left(\sum_{i=1}^k \exp(\beta \vec{z}_i^T \vec{x}) \right) + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

Apply Jensen's inequality again but with specific $p_i = \frac{1}{k}$

$$E \leq - \sum_{i=1}^k \frac{1}{k} \vec{z}_i^T \vec{x} + \frac{1}{2} \vec{x}^T \vec{x} + \beta^{-1} \ln k + \frac{1}{2} M^2$$

$$\leq - \sum_{i=1}^k \frac{1}{k} \vec{z}_i^T \vec{x} + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2$$

$$= - \left(\underbrace{\sum_{i=1}^k \frac{1}{k} \vec{z}_i^T}_{\text{average of patterns}} \right) \vec{x} + \frac{1}{2} \vec{x}^T \vec{x} + \frac{1}{2} M^2$$

$$\leq \left\| \sum_{i=1}^k \frac{1}{k} \vec{z}_i^T \right\| \|\vec{x}\| + \frac{1}{2} \|\vec{x}\|^2 + \frac{1}{2} M^2$$

$$\leq M \cdot M + \frac{1}{2} M \cdot M + \frac{1}{2} M^2$$

$$\Rightarrow M^2$$

Note $\vec{x} = \sum_i p_i \vec{z}_i$

$$\|\vec{x}\| = \left\| \sum_i p_i \vec{z}_i \right\| \leq \sum_i p_i \|\vec{z}_i\| \leq \sum_i p_i M = M$$